

'Electrodeless' Induction Lamps Last Longer

Void of electrodes, induction lamps can outlast 100 incandescent, five HID or five typical fluorescent lamp changes

■ T.A. BABU

With the world going crazy about 'green', lighting companies are also vying with each other to be in sync with the trend. Going 'green' is not only sensible and environmentally helpful, it is also cost-effective. Induction lighting is such a product that can reduce electricity consumption, has a life expectancy of 100,000 hours, and is more environment friendly than conventional lighting.

The induction lamp system is a revolutionary technology of light generation that combines the basic principles of electromagnetic induction and gas ionisation. It is one of the best technologies developed, yet many designers and lighting distributors in the industry are still not familiar with the product. The reasons for its lack of popularity are higher initial cost and the special designs that it requires. However, induction lamps are slowly catching up as these last considerably longer than high-intensity discharge (HID) or high-pressure sodium (HPS) lamps. With a virtually zero maintenance requirement, they offer substantial savings in direct maintenance costs, while eliminating the inconvenience and indirect costs due to service interruption for maintenance work. Void of electrodes, induction lamps system can outlast 100 incandescent, five HID or five typical fluorescent lamp changes.

Brains behind the system

The basic technology for wireless transfer of power to electrodeless fluo-



rescent lamps is not particularly new. It was conceived by Nikola Tesla way back in 1890. General Electric is the world's first company which brought the commercial light source to combine the electrodeless induction lamp and all associated electronic control gear into a single retrofit unit. Several years earlier, Matsushita and Philips were the first to market the induction lamp, but those lamps required an external ballast. Induction lighting system was never a breakthrough science, and until recently, it has not been so commercially viable.

Development of induction lamp involved decades of effort in researches on relevant gas-discharge physics, solidstate physics, material science and electronic ballast. New developments have broken down the barriers of costs and technological setbacks, such as EMC interference, lumen depreciation, ability to dim and a useful range of available wattages. Today, induction

lamps are becoming a clear-cut choice for many lighting applications over traditional light sources.

Technology at a glance

In electrodeless lamps, the power needed to generate light is transferred from the outside of the lamp envelope by means of electromagnetic field. Induction lamps use an induction coil to create a magnetic field inside a gas-electron/ion plasma. The gas generates ultraviolet (UV) light, which excites a

Salient Features

1. No electrodes or filaments
2. 60,000 hrs rated average life
3. Low lumen depreciation of < 30% @ 60,000 hours
4. High-frequency operation
5. High system efficacy (65-70 lm/W)
6. No flickering, no stroboscopic effect
7. Electronic control
8. Instant (re)-start
9. Choice of colour temperature and excellent colour rendition